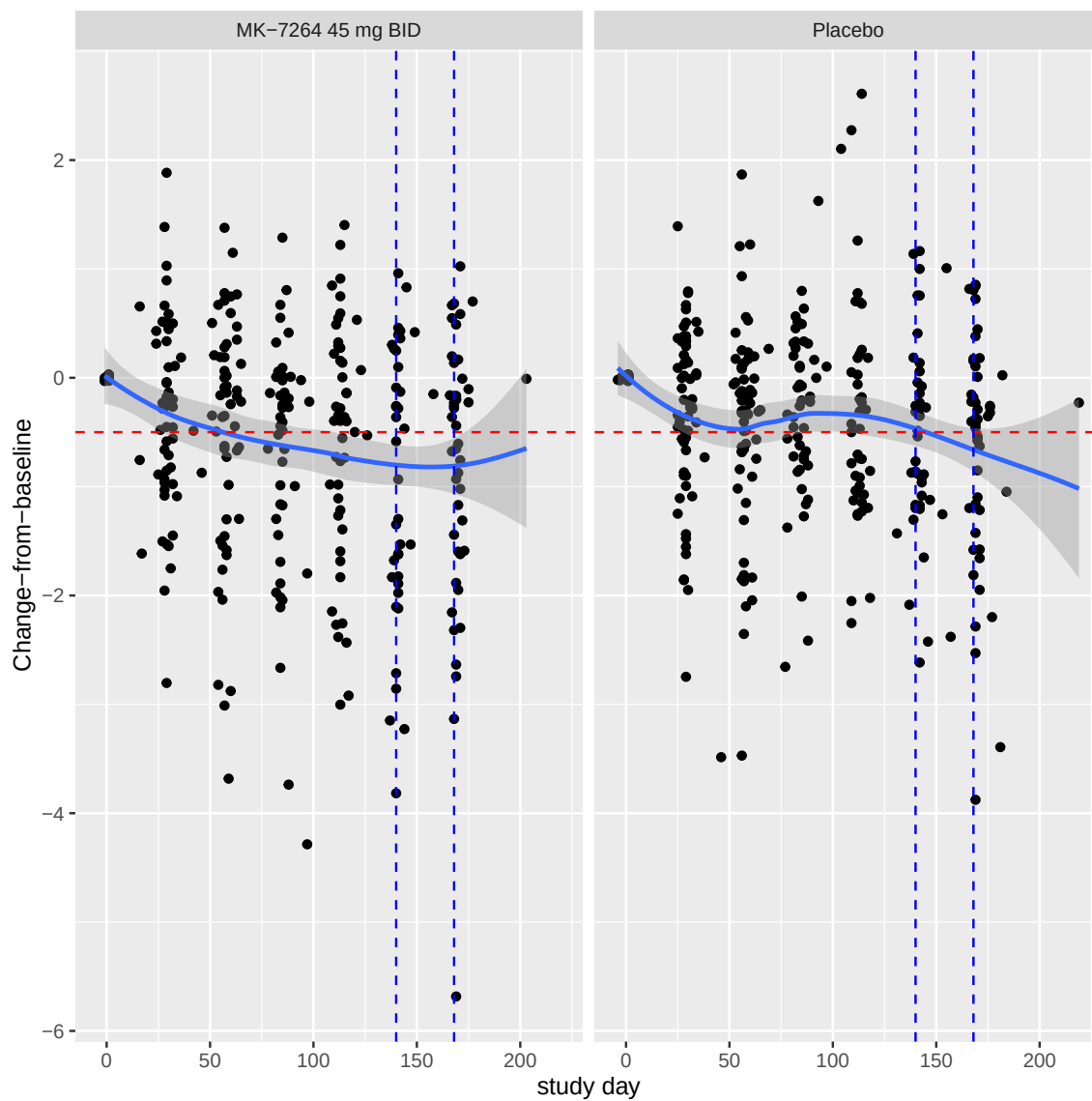


```
## Loading required package: leaps
## Loading required package: survival
## Loading required package: lattice
## Loading required package: Formula
## Loading required package: ggplot2
##
## Attaching package: 'Hmisc'
## The following objects are masked from 'package:base':
##
##      format.pval, units
## 'geom_smooth()' using method = 'loess' and formula 'y ~ x'
## Warning: Removed 8 rows containing non-finite values (stat_smooth).
## Warning: Removed 8 rows containing missing values (geom_point).
```



```

##
## N Y
## 5 92
##
## MK-7264 45 mg BID Placebo
## 0 46 51
## 1 19 14
##
##
## Descriptive Statistics by trtp
##
## +-----+-----+-----+-----+
## | |N |MK-7264 45 mg BID |Placebo |P-value |
## | | | (N=65) | (N=65) | |
## +-----+-----+-----+-----+
## |y_0 |130| 2.0/2.8/4.0 | 2.0/2.8/3.3 | 0.23 |
## +-----+-----+-----+-----+
## |sex : M |130| 51% (33) | 51% (33) | 1 |
## +-----+-----+-----+-----+
## |age |130| 37/50/61 | 31/43/57 | 0.14 |
## +-----+-----+-----+-----+
## |chg_4_impute|130| -0.84/-0.19/ 0.13 | -0.66/-0.20/ 0.20 | 0.83 |
## +-----+-----+-----+-----+
## |chg_8_impute|130| -0.91/-0.21/ 0.21 | -0.72/-0.30/ 0.11 | 0.8 |
## +-----+-----+-----+-----+
## |last_treat8 |130| 15% (10) | 3% ( 2) | 0.015|
## +-----+-----+-----+-----+
## |last_treat12|130| 22% (14) | 8% ( 5) | 0.025|
## +-----+-----+-----+-----+
## |last_treat16|130| 29% (19) | 15% (10) | 0.058|
## +-----+-----+-----+-----+
## |last_treat20|130| 31% (20) | 18% (12) | 0.1 |
## +-----+-----+-----+-----+
## |miss_24 |130| 29% (19) | 22% (14) | 0.31 |
## +-----+-----+-----+-----+
## |chg_24 | 97|-1.541/-0.411/-0.057|-1.202/-0.378/ 0.035| 0.81 |
## +-----+-----+-----+-----+
## |Hchg_24 | 92|-1.513/-0.373/-0.061|-1.204/-0.378/ 0.039| 0.91 |
## +-----+-----+-----+-----+
##
##
## Descriptive Statistics by trtp
##
## +-----+-----+-----+-----+
## | |N |MK-7264 45 mg BID |Placebo |P-value |
## | | | (N=65) | (N=65) | |
## +-----+-----+-----+-----+
## |y_0 |130| 2.0/2.8/4.0 | 2.0/2.8/3.3 | 0.23 |
## +-----+-----+-----+-----+
## |sex : M |130| 51% (33) | 51% (33) | 1 |
## +-----+-----+-----+-----+
## |age |130| 37/50/61 | 31/43/57 | 0.14 |
## +-----+-----+-----+-----+

```

```
## |chg_4_impute|130| -0.84/-0.19/ 0.13 | -0.66/-0.20/ 0.20 | 0.83 |
## +-----+-----+-----+-----+-----+
## |chg_8_impute|130| -0.91/-0.21/ 0.21 | -0.72/-0.30/ 0.11 | 0.8 |
## +-----+-----+-----+-----+
## |last_treat8 |130| 15% (10) | 3% ( 2) | 0.015|
## +-----+-----+-----+-----+
## |last_treat12|130| 22% (14) | 8% ( 5) | 0.025|
## +-----+-----+-----+-----+
## |last_treat16|130| 29% (19) | 15% (10) | 0.058|
## +-----+-----+-----+-----+
## |last_treat20|130| 31% (20) | 18% (12) | 0.1 |
## +-----+-----+-----+-----+
## |miss_20 |130| 40% (26) | 37% (24) | 0.72 |
## +-----+-----+-----+-----+
## |chg_20 | 80|-1.821/-0.599/ 0.172|-1.169/-0.511/-0.071| 0.23 |
## +-----+-----+-----+-----+
## |Hchg_20 | 78|-1.810/-0.599/ 0.074|-1.169/-0.511/-0.071| 0.23 |
## +-----+-----+-----+-----+
```

```
library(gee)
#library(tab)
library("kableExtra")
require(reshape2)

## Loading required package: reshape2

require(reshape)

## Loading required package: reshape
##
## Attaching package: 'reshape'
## The following objects are masked from 'package:reshape2':
##
##   colsplit, melt, recast

library(data.table)

##
## Attaching package: 'data.table'
## The following object is masked from 'package:reshape':
##
##   melt
## The following objects are masked from 'package:reshape2':
##
##   dcast, melt

#library("Matching")
#library("rms") # For MAR1 robust variance estimation

ep.name<-"Hchg_24"
or.names<-c("y_0", "age", "female", "y_8_impute")
ps.names<-c("y_0", "y_8_impute")
miss.name<-c("miss_H24")
treat.name<-c("treat_45mg")
direction.superiority<-c("negative") # Negative values in favor of experimental
```

```

dr.fit<-get.dr.analyses(df=df24,ep.name=ep.name,or.names=or.names,ps.names=ps.names,miss.name=miss.name
                      treat.name=treat.name,direction.superiority=direction.superiority,details=FALSE)

print(dr.fit$dr1.out[c("m0","m1","d","sed","dL","dU","pval.1sided")])

##           m0           m1           d           sed           dL           dU pval.1sided
## 1 -0.7154076 -0.6337209 0.08168672 0.2252481 -0.3597995 0.523173 0.6415677

print(dr.fit$dr2.out[c("m0","m1","d","sed","dL","dU","pval.1sided")])

##           m0           m1           d           sed           dL           dU pval.1sided
## 1 -0.6985613 -0.6290059 0.06955537 0.2252711 -0.3719761 0.5110868 0.6212491

ep.name<-"Hchg_20"
or.names<-c("y_0","age","female","y_8_impute")
ps.names<-c("y_0","y_8_impute")
miss.name<-c("miss_H20")
treat.name<-c("treat_45mg")
direction.superiority<-c("negative") # Negative values in favor of experimental

dr.fit<-get.dr.analyses(df=df20,ep.name=ep.name,or.names=or.names,ps.names=ps.names,miss.name=miss.name
                      treat.name=treat.name,direction.superiority=direction.superiority,details=FALSE)

print(dr.fit$dr1.out[c("m0","m1","d","sed","dL","dU","pval.1sided")])

##           m0           m1           d           sed           dL           dU pval.1sided
## 1 -0.5788211 -0.796534 -0.2177129 0.2061987 -0.6218624 0.1864365 0.1455205

print(dr.fit$dr2.out[c("m0","m1","d","sed","dL","dU","pval.1sided")])

##           m0           m1           d           sed           dL           dU pval.1sided
## 1 -0.5704877 -0.7872349 -0.2167472 0.2071319 -0.6227258 0.1892314 0.1476833

# Remove subjects 1 at a time to evaluate influence on DR estimator

ep.name<-"Hchg_24"
or.names<-c("y_0","age","female","y_8_impute")
ps.names<-c("y_0","y_8_impute")
miss.name<-c("miss_H24")
treat.name<-c("treat_45mg")
direction.superiority<-c("negative") # Negative values in favor of experimental

dr.fit<-get.dr.analyses(df=df24,ep.name=ep.name,or.names=or.names,ps.names=ps.names,miss.name=miss.name
                      treat.name=treat.name,direction.superiority=direction.superiority,details=FALSE)

#print(dr.fit$dr1.out[c("m0","m1","d","sed","dL","dU","pval.1sided")])
#print(dr.fit$dr2.out[c("m0","m1","d","sed","dL","dU","pval.1sided")])

dr1_out_delete<-matrix(NA,nrow=nrow(df24),ncol=9)
dr2_out_delete<-matrix(NA,nrow=nrow(df24),ncol=9)

for(ii in 1:nrow(df24)){
df24_d<-df24[-ii,]

dr_d<-get.dr.analyses(df=df24_d,ep.name=ep.name,or.names=or.names,ps.names=ps.names,miss.name=miss.name

```

```

        treat.name=treat.name,direction.superiority=direction.superiority,details=FALSE)

temp <- as.numeric(c(ii,c(df24$subjid[ii]),dr_d$dr1.out[c("m0","m1","d","sed","dL","dU","pval.1sided")]))
dr1_out_delete[ii,] <- temp

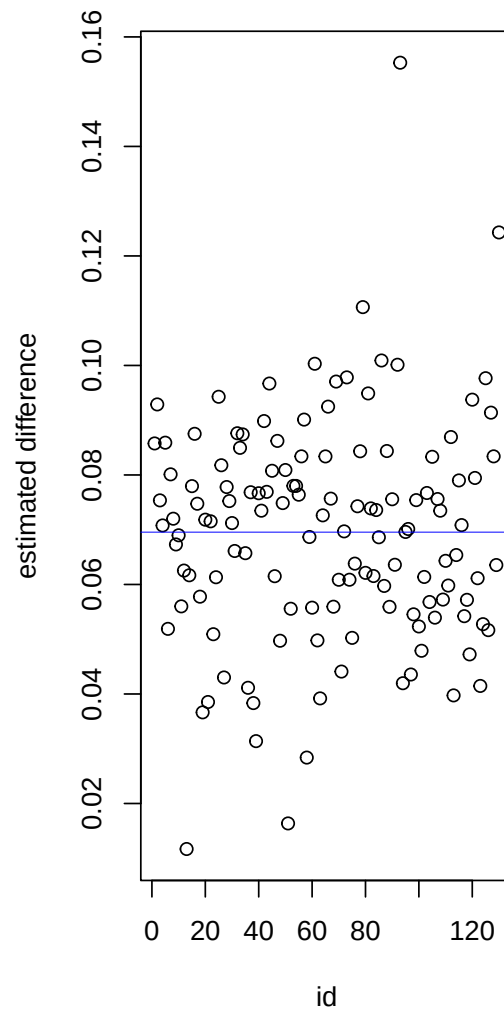
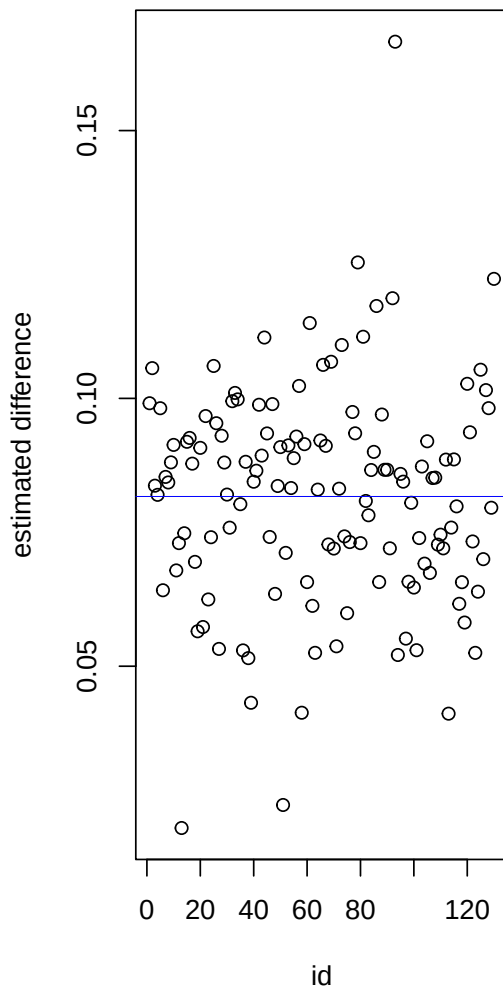
temp <- as.numeric(c(ii,c(df24$subjid[ii]),dr_d$dr2.out[c("m0","m1","d","sed","dL","dU","pval.1sided")]))
dr2_out_delete[ii,] <- temp
}
dr1_out_delete <- data.table(dr1_out_delete)
dr2_out_delete <- data.table(dr2_out_delete)

names(dr1_out_delete)<-c("id_n","subjid","m0","m1","d","sed","dL","dU","pval")
names(dr2_out_delete)<-c("id_n","subjid","m0","m1","d","sed","dL","dU","pval")

par(mfrow=c(1,2))
with(dr1_out_delete,plot(id_n,d,xlab="id",ylab="estimated difference"))
abline(h=c(dr.fit$dr1.out$d),lwd=0.5,lty=1,col="blue")

with(dr2_out_delete,plot(id_n,d,xlab="id",ylab="estimated difference"))
abline(h=c(dr.fit$dr2.out$d),lwd=0.5,lty=1,col="blue")

```



```
# Subjects that reduce point estimate by more than 20%
```

```
dr1_out_delete$d_all<-c(dr.fit$dr1.out$d)
dr1_out_delete<-within(dr1_out_delete,{
  pct_delD<-(d-d_all)/d_all
})
dr2_out_delete$d_all<-c(dr.fit$dr2.out$d)
dr2_out_delete<-within(dr2_out_delete,{
  pct_delD<-(d-d_all)/d_all
})
```

```
# Subjects --- when removed -- with 40% improvement
```

```
dr1_best<-subset(dr1_out_delete,pct_delD<=-0.4)
dr2_best<-subset(dr2_out_delete,pct_delD<=-0.4)
print(dr1_best)
```

```
##      id_n subjid      m0      m1      d      sed      dL      dU
## 1:    13 402448 -0.6535146 -0.6337209 0.01979375 0.2192361 -0.4099090 0.4494965
## 2:    39 412466 -0.7154076 -0.6722338 0.04317378 0.2234690 -0.3948255 0.4811731
## 3:    51 412483 -0.6578104 -0.6337209 0.02408952 0.2201879 -0.4074788 0.4556578
## 4:    58 414805 -0.6750331 -0.6337209 0.04131222 0.2235180 -0.3967830 0.4794075
## 5:   113 415022 -0.7154076 -0.6742669 0.04114066 0.2226839 -0.3953199 0.4776012
##      pval      d_all      pct_delD
## 1: 0.5359697 0.08168672 -0.7576870
## 2: 0.5765981 0.08168672 -0.4714713
## 3: 0.5435591 0.08168672 -0.7050987
## 4: 0.5733177 0.08168672 -0.4942602
## 5: 0.5732871 0.08168672 -0.4963605
```

```
print(dr2_best)
```

```
##      id_n subjid      m0      m1      d      sed      dL      dU
## 1:    13 402448 -0.6406966 -0.6290059 0.01169072 0.2189100 -0.4173729 0.4407544
## 2:    19 402458 -0.6656491 -0.6290059 0.03664322 0.2245010 -0.4033788 0.4766652
## 3:    21 402460 -0.6985613 -0.6600257 0.03853552 0.2262289 -0.4048732 0.4819442
## 4:    36 412462 -0.6701397 -0.6290059 0.04113384 0.2255744 -0.4009919 0.4832596
## 5:    38 412465 -0.6985613 -0.6602197 0.03834160 0.2251154 -0.4028845 0.4795677
## 6:    39 412466 -0.6985613 -0.6671724 0.03138889 0.2236930 -0.4070494 0.4698271
## 7:    51 412483 -0.6453688 -0.6290059 0.01636289 0.2198424 -0.4145283 0.4472541
## 8:    58 414805 -0.6573954 -0.6290059 0.02838953 0.2237017 -0.4100658 0.4668449
## 9:    63 414812 -0.6681996 -0.6290059 0.03919370 0.2255707 -0.4029249 0.4813123
## 10:   113 415022 -0.6985613 -0.6588198 0.03974143 0.2255745 -0.4023846 0.4818674
## 11:   123 415032 -0.6704782 -0.6290059 0.04147231 0.2258100 -0.4011152 0.4840598
##      pval      d_all      pct_delD
## 1: 0.5212951 0.06955537 -0.8319221
## 2: 0.5648277 0.06955537 -0.4731792
## 3: 0.5676281 0.06955537 -0.4459735
## 4: 0.5723466 0.06955537 -0.4086173
## 5: 0.5676207 0.06955537 -0.4487615
## 6: 0.5557969 0.06955537 -0.5487209
## 7: 0.5296659 0.06955537 -0.7647501
## 8: 0.5504934 0.06955537 -0.5918428
## 9: 0.5689704 0.06955537 -0.4365108
## 10: 0.5699232 0.06955537 -0.4286361
## 11: 0.5728600 0.06955537 -0.4037512
```

```
#      id_n subjid      m0      m1      d      sed      dL      dU      pval      d_all
#1:    13 402448 -0.6535146 -0.6337209 0.01979375 0.2192361 -0.4099090 0.4494965 0.5359697 0.08168672
#2:    39 412466 -0.7154076 -0.6722338 0.04317378 0.2234690 -0.3948255 0.4811731 0.5765981 0.08168672
#3:    51 412483 -0.6578104 -0.6337209 0.02408952 0.2201879 -0.4074788 0.4556578 0.5435591 0.08168672
#4:    58 414805 -0.6750331 -0.6337209 0.04131222 0.2235180 -0.3967830 0.4794075 0.5733177 0.08168672
#5:   113 415022 -0.7154076 -0.6742669 0.04114066 0.2226839 -0.3953199 0.4776012 0.5732871 0.08168672
```

```
# What if we remove from analysis
```

```
subs_toremove<-dr1_best$subjid
```

```
# Just loop through
```

```
df24_new<-df24
```

```
df24_new$mark<-0.0
```

```

for(jj in 1:length(subs_toremove)){
  id_loc<-which(df24_new$subjid==subs_toremove[jj])
  df24_new$mark[id_loc]<-1.0
}

df24_OptSubset<-subset(df24_new,mark==0)

bl_24<-summary(trtp ~ y_0+sex+age+chg_4_impute+chg_8_impute+last_treat8+last_treat12+last_treat16+last_
print(bl_24,digits=2,prtest=c('P'))

##
##
## Descriptive Statistics by trtp
##
## +-----+-----+-----+-----+-----+
## |           |N |MK-7264 45 mg BID |Placebo           |P-value           |
## |           | | |(N=63)           | |(N=62)           | |
## +-----+-----+-----+-----+-----+
## |y_0         |125| 2.1/2.8/4.0 | 2.0/2.8/3.4 | 0.24 |
## +-----+-----+-----+-----+-----+
## |sex : M     |125| 52% (33) | 50% (31) | 0.79 |
## +-----+-----+-----+-----+-----+
## |age         |125| 37/50/62 | 32/44/57 | 0.22 |
## +-----+-----+-----+-----+-----+
## |chg_4_impute|125| -0.85/-0.20/ 0.05 | -0.65/-0.20/ 0.19 | 0.65 |
## +-----+-----+-----+-----+-----+
## |chg_8_impute|125| -0.95/-0.25/ 0.15 | -0.69/-0.29/ 0.14 | 0.86 |
## +-----+-----+-----+-----+-----+
## |last_treat8 |125| 16% (10) | 3% ( 2) | 0.016|
## +-----+-----+-----+-----+-----+
## |last_treat12|125| 22% (14) | 8% ( 5) | 0.027|
## +-----+-----+-----+-----+-----+
## |last_treat16|125| 29% (18) | 16% (10) | 0.095|
## +-----+-----+-----+-----+-----+
## |last_treat20|125| 30% (19) | 19% (12) | 0.16 |
## +-----+-----+-----+-----+-----+
## |miss_24     |125| 29% (18) | 23% (14) | 0.44 |
## +-----+-----+-----+-----+-----+
## |chg_24      | 93|-1.570/-0.448/-0.125|-1.139/-0.350/ 0.055| 0.36 |
## +-----+-----+-----+-----+-----+
## |Hchg_24     | 88|-1.541/-0.411/-0.127|-1.162/-0.350/ 0.087| 0.42 |
## +-----+-----+-----+-----+-----+

dfa<-df24_OptSubset

ep.name<-"Hchg_24"
or.names<-c("y_0","age","female","y_8_impute")
ps.names<-c("y_0","y_8_impute")
miss.name<-c("miss_H24")
treat.name<-c("treat_45mg")
direction.superiority<-c("negative") # Negative values in favor of experimental

dr.fit<-get.dr.analyses(df=dfa,ep.name=ep.name,or.names=or.names,ps.names=ps.names,miss.name=miss.name,
treat.name=treat.name,direction.superiority=direction.superiority,details=FALSE)

```



```

print(dr.fit$dr1.out[c("m0", "m1", "d", "sed", "dL", "dU", "pval.1sided")])

##           m0           m1           d           sed           dL           dU pval.1sided
## 1 -0.548514 -0.710894 -0.16238 0.2066051 -0.5673261 0.2425661 0.2159503

print(dr.fit$dr2.out[c("m0", "m1", "d", "sed", "dL", "dU", "pval.1sided")])

##           m0           m1           d           sed           dL           dU pval.1sided
## 1 -0.5400337 -0.6965972 -0.1565635 0.2086299 -0.5654781 0.2523511 0.2264959

# Who?

df_who <- subset(df24_new, mark==1)
print(df_who)

##      subjid      y_0 age sex      trtp treat_45mg offtrt_day      y_4
## 13  402448  3.0583  31  M      Placebo           0          363      NA
## 39  412466  3.6889  48  F MK-7264 45 mg BID           1          360  4.1865
## 51  412483  1.3652  24  F      Placebo           0          363  2.7752
## 58  414805  3.1867  48  M      Placebo           0          386  1.6422
## 113 415022  2.0093  57  F MK-7264 45 mg BID           1          107  3.8712
##      chg_ontrt_4      y_8 chg_ontrt_8 chg_8_impute chg_4_impute y_8_impute
## 13           NA           NA           NA      0.0000      0.0000      3.0583
## 39  0.4976137  3.9020  0.2130932      0.2131      0.4976      3.9020
## 51  1.4099486 -2.0794 -3.4446825     -3.4446      1.4100     -2.0794
## 58  -1.5444692           NA           NA     -1.5445     -1.5445      1.6422
## 113  1.8618690  3.3615  1.3522001      1.3522      1.8619      3.3615
##      y_4_impute miss_4_8      y_24 chg_ontrt_24 ontrtfl_24 chg_24 Hchg_24 Hy_24
## 13      3.0583           1 -0.3448     -3.403156           Y -3.4031 -3.4031 -0.3448
## 39      4.1865           0  4.7325      1.043657           Y  1.0436  1.0436  4.7325
## 51      2.7752           0 -2.4849     -3.850148           Y -3.8501 -3.8501 -2.4849
## 58      1.6422           0  0.6380     -2.548716           Y -2.5487 -2.5487  0.6380
## 113      3.8712           0      NA           NA      <NA>      NA      NA      NA
##      female miss_H24 miss_24 miss_offtreat last_treat20 last_treat16
## 13           0           0           0           0           0           0
## 39           1           0           0           0           0           0
## 51           1           0           0           0           0           0
## 58           0           0           0           0           0           0
## 113          1           1           1           0           1           1
##      last_treat12 last_treat8 offtreat_day mark
## 13              0              0          168    1
## 39              0              0          168    1
## 51              0              0          168    1
## 58              0              0          168    1
## 113             0              0          107    1

print(df_who$subjid)

## [1] 402448 412466 412483 414805 415022

```

```

# Longitudinal data
dflong_new <- df_long
dflong_new$mark <- 0.0
for(jj in 1:length(subs_toremove)){

```

```

id_loc<-which(dflong_new$subjid==subs_toremove[jj])
dflong_new$mark[id_loc]<-1.0
}

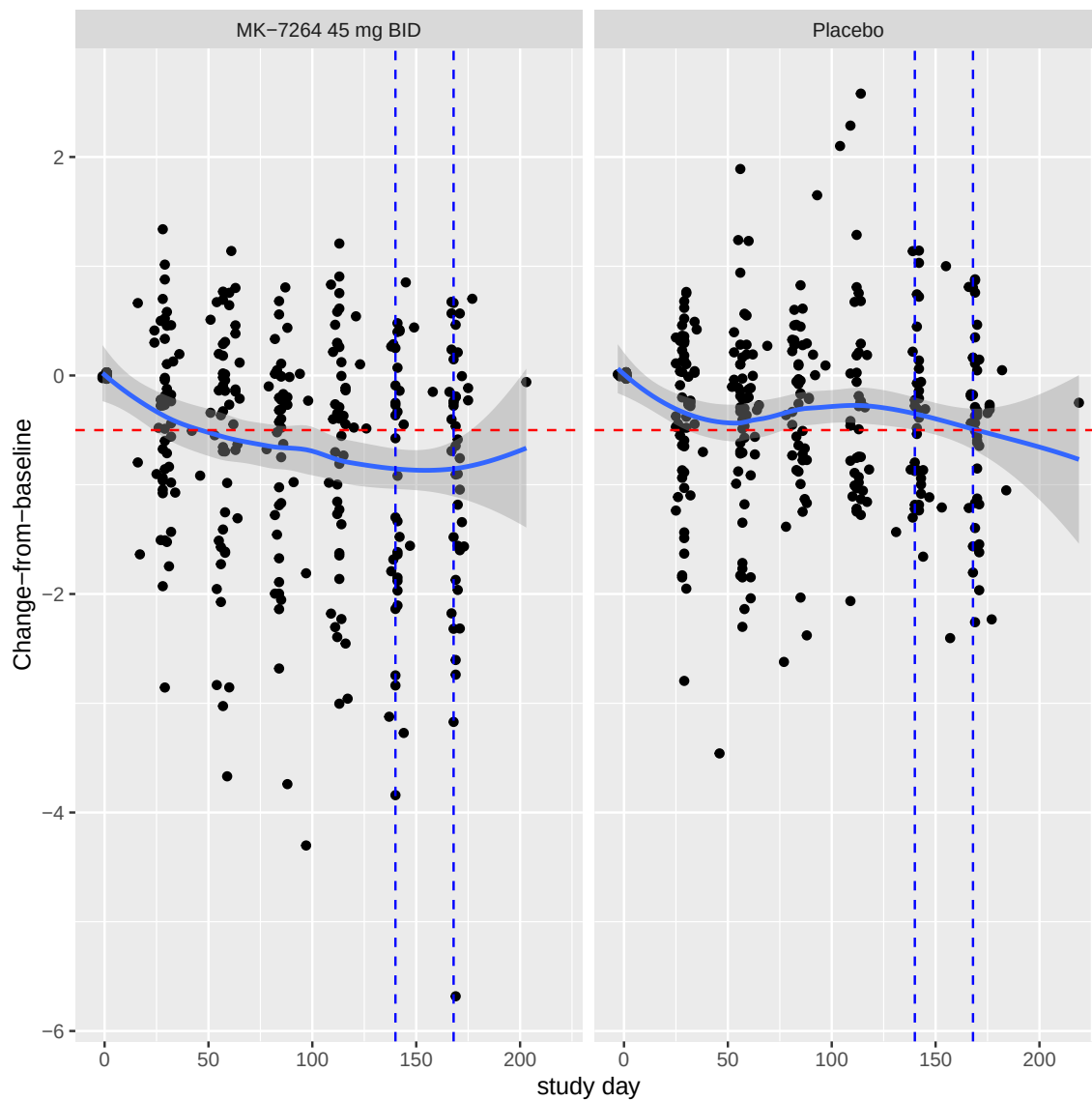
dflong_OptSubset<-subset(dflong_new,mark==0)

# Plot longitudinal
p<-ggplot(data=dflong_OptSubset,aes(x=ady,y=chg,group=subjid))
p<-p+xlab("study day")+ylab("Change-from-baseline")
p<-p+geom_point(position=position_jitter(height=0.03,width=0))+stat_smooth(aes(group=1))+facet_grid(. ~
p<-p+geom_hline(yintercept=-0.50,linetype="dashed",color="red")
p<-p+geom_vline(xintercept=c(140,168),linetype="dashed",color="blue")

plot(p)

## 'geom_smooth()' using method = 'loess' and formula 'y ~ x'
## Warning: Removed 8 rows containing non-finite values (stat_smooth).
## Warning: Removed 8 rows containing missing values (geom_point).

```



```
# Just loop through
df20_new<-df20
df20_new$mark<-0.0
for(jj in 1:length(subs_toremove)){
  id_loc<-which(df20_new$subjid==subs_toremove[jj])
  df20_new$mark[id_loc]<-1.0
}

df20_OptSubset<-subset(df20_new,mark==0)

dfa<-df20_OptSubset

ep.name<-"Hchg_20"
or.names<-c("y_0","age","female","y_8_impute")
ps.names<-c("y_0","y_8_impute")
miss.name<-c("miss_H20")
```

```

treat.name<-c("treat_45mg")
direction.superiority<-c("negative") # Negative values in favor of experimental

dr.fit<-get.dr.analyses(df=dfa,ep.name=ep.name,or.names=or.names,ps.names=ps.names,miss.name=miss.name,
                        treat.name=treat.name,direction.superiority=direction.superiority,details=FALSE)

print(dr.fit$dr1.out[c("m0","m1","d","sed","dL","dU","pval.1sided")])

##           m0           m1           d           sed           dL           dU pval.1sided
## 1 -0.44817 -0.8670085 -0.4188384 0.1919499 -0.7950603 -0.04261655 0.01455406

print(dr.fit$dr2.out[c("m0","m1","d","sed","dL","dU","pval.1sided")])

##           m0           m1           d           sed           dL           dU pval.1sided
## 1 -0.4388666 -0.8541429 -0.4152763 0.1938876 -0.7952961 -0.03525655 0.01610317

```